SQL Sales Performance Analysis Using MySQL

End-to-End SQL Data Analytics Project Report

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Academic Program:	B.Tech in Artificial Intelligence and Data Science

Project Type:	Data Analytics Portfolio Project
Dataset:	Superstore Sales Dataset
Database:	sales_analysis
Table:	superstore_sales
Tools Used:	MySQL Workbench, SQL, GitHub

June 30, 2026

Certificate

This is to certify that **Sumit Kumar**, a student of **B.Tech in Artificial Intelligence and Data Science**, has prepared the project report titled **"SQL Sales Performance Analysis Using MySQL"**. The project is based on an end-to-end SQL data analytics workflow performed using MySQL Workbench on the Superstore Sales Dataset.

The project demonstrates practical knowledge of SQL querying, data validation, KPI calculation, sales performance analysis, product profitability analysis, customer segment analysis, regional comparison, and business insight generation. The work is suitable for academic learning, GitHub portfolio presentation, LinkedIn showcase, internship applications, and entry-level data analyst interview discussion.

Declaration

I, **Sumit Kumar**, a B.Tech student specializing in Artificial Intelligence and Data Science, hereby declare that this project report titled **“SQL Sales Performance Analysis Using MySQL”** has been prepared as an original data analytics portfolio project. The analysis has been completed using SQL in MySQL Workbench on the Superstore Sales Dataset.

The report presents my understanding of database-driven analysis, business intelligence, KPI reporting, and analytical storytelling. The insights, interpretations, and recommendations are based on the SQL query results generated from the dataset. This report has been prepared for learning, professional portfolio development, and demonstration of entry-level data analyst skills.

Name: Sumit Kumar
Program: B.Tech in Artificial Intelligence and Data Science
Date: June 30, 2026

Acknowledgement

I would like to express my sincere gratitude to my teachers, mentors, and learning communities who helped me develop the technical foundation required to complete this SQL data analytics project. I am also thankful for the availability of the Superstore Sales Dataset, which provides a realistic business case for practicing sales analysis, profitability analysis, and business intelligence reporting.

This project helped me strengthen my understanding of SQL, MySQL Workbench, data cleaning, data validation, aggregation, filtering, grouping, ranking, and business interpretation. It also improved my ability to present analytical results in a structured professional report format suitable for recruiters, interviewers, and portfolio reviewers.

Abstract

This project report presents an end-to-end SQL data analytics study titled **“SQL Sales Performance Analysis Using MySQL”**. The analysis was conducted on the Superstore Sales Dataset using MySQL Workbench. The dataset contains **9,694 rows** and **21 columns**, covering sales transactions from **2014 to 2017**. The data was cleaned and validated before analysis, with no missing values and no duplicate Row IDs identified.

The objective of this project is to analyze business performance using SQL and convert raw transactional data into meaningful business insights. The project calculates important business KPIs such as total sales, total profit, total quantity sold, and total orders. It also examines product profitability, loss-making products, category performance, sub-category performance, regional and state-level sales, city-wise performance, customer segments, shipping modes, and year-wise trends.

The analysis found total sales of **$2,272,449.86**, total profit of **$286,857.75**, total quantity sold of **36,749**, and **4,931** total orders. Technology was the strongest category by both sales and profit, while Furniture showed weak profitability despite high sales. The West region generated the highest sales and profit, and the Consumer segment contributed the largest share of revenue and profit. The report concludes with business insights and recommendations focused on profitability improvement, discount control, regional strategy, product optimization, and customer targeting.

Contents

1 Introduction

Data analytics plays an important role in modern business decision-making. Organizations generate large volumes of transactional data through sales, customer interactions, shipping activities, and product purchases. When this data is analyzed correctly, it can reveal valuable patterns related to revenue, profit, customer behavior, product performance, and operational efficiency.

This project focuses on analyzing the Superstore Sales Dataset using SQL in MySQL Workbench. The dataset represents retail sales transactions across different products, categories, customer segments, cities, states, regions, and shipping modes. The purpose of the project is to demonstrate how SQL can be used to perform structured business analysis and generate insights that support decision-making.

As a B.Tech student in Artificial Intelligence and Data Science, this project also supports the development of practical data analytics skills. It connects database querying with business interpretation, which is essential for data analyst roles. The report is written in a professional format so it can be used as a GitHub portfolio project, LinkedIn showcase, internship submission, and interview discussion document.

2 Problem Statement

The Superstore business has transactional sales data, but raw data alone does not provide direct answers to business questions. Decision makers need clear summaries of sales, profit, product performance, customer behavior, regional contribution, and yearly trends. Without structured analysis, it becomes difficult to identify profitable products, loss-making areas, strong regions, weak cities, and opportunities for growth.

The problem addressed in this project is:

> *How can SQL be used to analyze Superstore sales data and identify key business performance patterns, profitability drivers, loss areas, and strategic improvement opportunities?*

This project solves the problem by using SQL queries to validate the dataset, calculate KPIs, group data by important business dimensions, rank performance, identify losses, and generate professional business recommendations.

3 Project Objectives

The main objective of this project is to perform business intelligence and sales analytics using SQL. The specific objectives are:

- Analyze overall business performance using SQL.
- Calculate important business KPIs including sales, profit, quantity, and orders.
- Identify top-performing and loss-making products.
- Analyze category and sub-category performance.
- Analyze regional, state-level, and city-wise business performance.
- Evaluate customer segment contribution to sales and profit.
- Analyze shipping method performance.
- Study year-wise sales and profit trends from 2014 to 2017.
- Generate business insights and recommendations based on query results.

4 Tools and Technologies

Table 4.1: Tools and Technologies Used

Tool / Technology	**Purpose**
MySQL Workbench	Used for creating the database, importing the dataset, executing SQL queries, and reviewing query results.
SQL	Used for data validation, aggregation, grouping, filtering, ranking, and business analysis.
GitHub	Used for project documentation, version control, and portfolio presentation.
Superstore Sales Dataset	Used as the source data for retail sales and profit analysis.

SQL was selected because it is one of the most important tools for data analysts. It allows analysts to work directly with structured databases and produce reliable, repeatable business summaries. In this project, SQL was used to convert transaction-level data into performance metrics and business insights.

5 Project Methodology

The methodology followed in this project was designed to convert raw sales transaction data into meaningful business insights using SQL. The complete workflow included dataset collection, database setup, data import, cleaning, validation, KPI analysis, exploratory analysis, business analysis, insight generation, and recommendation development.

5.1 Textual Workflow Representation

```
Dataset Collection
        |
        v
Data Import into MySQL
        |
        v
Database Creation
        |
        v
Data Cleaning
        |
        v
Data Validation
        |
        v
KPI Analysis
        |
        v
Exploratory Data Analysis
        |
        v
```

```
Business Analysis
        |
        v
Business Insights
        |
        v
Business Recommendations
        |
        v
Conclusion
```

5.2 Dataset Collection

The first step of the project was collecting the Superstore Sales Dataset. This dataset contains sales transaction records related to orders, products, customers, geographic locations, shipping modes, sales values, quantities, discounts, and profit.

This step is important because the quality and relevance of the dataset directly affect the accuracy of the analysis. A business-oriented dataset was selected so that SQL queries could be used to answer practical sales and profitability questions.

5.3 Data Import into MySQL

After collecting the dataset, the data was imported into MySQL Workbench for structured analysis. The dataset was loaded into a relational database table so that SQL queries could be executed efficiently.

Importing the data into MySQL was important because SQL provides a reliable environment for filtering, grouping, aggregating, sorting, and analyzing structured data. It also makes the analysis repeatable and easier to document.

5.4 Database Creation

A database named **sales_analysis** was created to store and manage the project data. The main table used for analysis was named **superstore_sales**.

Creating a separate database helped organize the project properly and ensured that all analysis was performed in a dedicated workspace. This is an important practice in data analytics projects because it improves clarity, maintainability, and reproducibility.

5.5 Data Cleaning

Data cleaning was performed to ensure that the dataset was ready for analysis. This included checking the structure of the table, reviewing column names, verifying data types, and ensuring that the records were properly imported.

Data cleaning is important because raw datasets may contain formatting issues, incorrect values, duplicate records, or missing information. Clean data helps prevent incorrect calculations and supports more reliable business conclusions.

5.6 Data Validation

After cleaning, the dataset was validated by checking the total number of rows, total number of columns, missing values, duplicate Row IDs, and date range. The dataset contained **9,694 rows**, **21 columns**, no missing values, and no duplicate Row IDs.

Data validation is important because it confirms whether the dataset is complete, consistent, and trustworthy. Before generating KPIs or business insights, it is necessary to ensure that the underlying data is accurate.

5.7 KPI Analysis

Key Performance Indicators were calculated using SQL queries. The main KPIs included total sales, total profit, total quantity sold, and total orders. These KPIs provided a high-level summary of overall business performance.

KPI analysis is important because it gives stakeholders a quick understanding of the business position. It helps answer essential questions such as how much revenue was generated, how profitable the business was, and how many orders were processed.

5.8 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand patterns and trends in the dataset. The data was analyzed across categories, sub-categories, products, regions, states, cities, customer segments, shipping modes, and years.

This step is important because it helps identify areas that require deeper investigation. EDA allows analysts to discover performance differences, unusual trends, profitable segments, and loss-making areas.

5.9 Business Analysis

Business analysis was performed by interpreting SQL query results from a decision-making perspective. The analysis focused on identifying profitable products, loss-making products, strong categories, weak categories, top-performing regions, loss-making cities, customer segment contribution, shipping performance, and year-wise growth.

This step is important because data analysis becomes valuable only when it is connected to business meaning. Business analysis converts numbers into practical understanding that can support strategy and operations.

5.10 Business Insights

After completing the analysis, key insights were generated from the results. These insights highlighted important findings such as Technology being the most profitable category, Furniture having weak profitability, the West region leading in sales and profit, and Philadelphia showing high sales but negative profitability.

Business insights are important because they summarize the main lessons from the analysis. They help decision makers understand what is working well, what is underperforming, and where action is required.

5.11 Business Recommendations

Based on the insights, business recommendations were developed. These recommendations included promoting high-profit products, reviewing loss-making products, improving Furniture profitability, investigating loss-making cities, strengthening Consumer segment campaigns, and monitoring shipping mode profitability.

Recommendations are important because they convert analysis into action. A professional data analytics project should not only describe what happened but also suggest what the business should do next.

5.12 Conclusion

The methodology followed in this project provided a complete SQL-based analytics workflow, starting from dataset collection and ending with business recommendations. Each step helped transform raw Superstore sales data into structured insights.

This methodology demonstrates practical data analyst skills, including database creation, data cleaning, validation, SQL querying, KPI reporting, exploratory analysis, business interpretation, and recommendation writing.

6 Dataset Description

The project uses the Superstore Sales Dataset, which contains retail transaction records. Each row represents a sales transaction or order line item. The dataset includes fields related to order details, customer information, product details, geography, shipping method, sales value, quantity, discount, and profit.

Table 6.1: Dataset Summary

Attribute	Details
Dataset Name	Superstore Sales Dataset
Database Name	sales_analysis
Table Name	superstore_sales
Total Rows	9,694
Total Columns	21
Date Range	2014–2017
Missing Values	No missing values
Duplicate Row IDs	No duplicate Row IDs
Cleaning Status	Dataset successfully cleaned and validated

6.1 Key Business Dimensions

The dataset supports analysis across several important business dimensions:

- Product dimensions such as category, sub-category, and product name.
- Geographic dimensions such as region, state, and city.
- Customer dimensions such as customer segment.

- Shipping dimensions such as ship mode.
- Time dimensions such as order year.
- Business measures such as sales, profit, quantity, and order count.

7 Database Structure

The database used in this project is named **sales_analysis**. The primary table used for analysis is named **superstore_sales**. The table stores the complete Superstore dataset in a flat analytical format.

Table 7.1: Database Structure

Component	**Description**
Database	sales_analysis
Main Table	superstore_sales
Data Granularity	One row per order line item
Primary Identifier	Row ID
Main Numeric Fields	Sales, Profit, Quantity, Discount
Main Categorical Fields	Category, Sub-Category, Product Name, Segment, Region, State, City, Ship Mode
Main Date Field	Order Date

The flat table structure is suitable for this portfolio project because it allows direct querying, filtering, grouping, and aggregation. For larger production systems, this structure could be normalized into fact and dimension tables to support scalable reporting.

8 Database Schema and Data Model

This section explains the database schema and data model used in the project **"SQL Sales Performance Analysis Using MySQL"**. The project uses a MySQL database named **sales_analysis** and a primary analytical table named **superstore_sales**. The table stores transaction-level sales records from the Superstore Sales Dataset and supports analysis across products, customers, geography, shipping, dates, sales, quantity, discount, and profit.

8.1 Database Design

The database was designed as a single analytical database for SQL-based sales performance analysis. The database name used in this project is:

```
sales_analysis
```

Inside this database, the complete dataset was stored in one main table:

```
superstore_sales
```

The purpose of this design was to keep the project simple, clear, and suitable for portfolio-level SQL analysis. Since the dataset is already structured and contains all required business fields in one table, a flat analytical table was selected for direct querying and reporting.

This design allows SQL queries to be written efficiently for KPI calculation, product analysis, regional analysis, customer segment analysis, shipping analysis, and trend analysis.

8.2 Table Structure

The **superstore_sales** table contains 21 columns. These columns represent order details, customer details, geographic information, product details, and business measures.

Table 8.1: Structure of the superstore_sales Table

Column Name	Description
Row ID	Unique row-level identifier for each record in the dataset.
Order ID	Identifier for each customer order. One order may contain multiple product line items.
Order Date	Date on which the order was placed.
Ship Date	Date on which the order was shipped.
Ship Mode	Shipping method used for the order, such as Standard Class or First Class.
Customer ID	Unique identifier assigned to each customer.
Customer Name	Name of the customer who placed the order.
Segment	Customer segment, such as Consumer, Corporate, or Home Office.
Country	Country where the order was placed.
City	City associated with the order.
State	State associated with the order.
Postal Code	Postal code of the customer or order location.
Region	Geographic region associated with the order.
Product ID	Unique identifier assigned to each product.
Category	Main product category, such as Technology, Furniture, or Office Supplies.
Sub-Category	Detailed product group within the main category.
Product Name	Name of the product sold.
Sales	Revenue generated from the transaction line item.
Quantity	Number of units sold in the transaction line item.
Discount	Discount applied to the transaction line item.
Profit	Profit or loss generated from the transaction line item.

8.3 Primary Identifier

The primary row-level identifier in the dataset is **Row ID**. Each Row ID represents a unique record in the **superstore_sales** table.

Although **Order ID** is also an important business identifier, it is not unique at the row level because a single order may include multiple products. Therefore, **Row ID** is treated as the primary identifier for record-level validation and duplicate checking.

8.4 Data Granularity

The granularity of the **superstore_sales** table is:

> *One row represents one product line item within a customer order.*

This means that if one customer order contains multiple products, the same Order ID may appear in multiple rows. Each row captures the sales, quantity, discount, and profit for a specific product line item.

Understanding data granularity is important because it affects how metrics are calculated. For example, total sales and total profit can be calculated by summing row-level values, while total orders should be calculated using distinct Order IDs.

8.5 Categorical Fields

Categorical fields are descriptive columns used for grouping, filtering, and segmentation. These fields help analyze business performance across different dimensions.

The main categorical fields in the table include:

- **Order ID**
- **Ship Mode**
- **Customer ID**
- **Customer Name**
- **Segment**

- **Country**
- **City**
- **State**
- **Postal Code**
- **Region**
- **Product ID**
- **Category**
- **Sub-Category**
- **Product Name**

These fields are used to answer questions such as which category is most profitable, which region has the highest sales, which cities are loss-making, and which customer segment contributes most to revenue.

8.6 Numerical Fields

Numerical fields are measurable values used for calculations, aggregations, and KPI reporting. The key numerical fields in the dataset are:

- **Sales**
- **Quantity**
- **Discount**
- **Profit**

These fields are used to calculate total sales, total profit, total quantity sold, average discount, product profitability, category performance, and regional performance.

For example, **Sales** represents revenue, while **Profit** represents the financial gain or loss from each transaction line item. **Quantity** measures sales volume, and **Discount** helps evaluate the impact of price reductions on profitability.

8.7 Date Fields

The dataset contains two important date fields:

- **Order Date**
- **Ship Date**

Order Date is used to analyze sales and profit trends over time. It supports year-wise, month-wise, and quarter-wise analysis. **Ship Date** can be used to analyze shipping timelines and delivery performance.

Date fields are important because they allow the business to understand growth patterns, seasonal trends, and operational delays.

8.8 Reason for Choosing a Flat Analytical Table

A flat analytical table was chosen because the Superstore dataset is already organized in a single structured table. For this SQL portfolio project, the flat table design provides a simple and effective way to perform analysis without requiring complex joins.

This approach is suitable because:

- All required fields are available in one table.
- SQL queries are easier to write and understand.
- KPI calculations can be performed directly.
- It is suitable for beginner-to-intermediate analytics reporting.
- It supports fast exploration of sales, profit, customer, product, and geographic dimensions.

The flat table structure is practical for portfolio demonstration because it allows the focus to remain on SQL analysis, business interpretation, and insight generation.

8.9 Advantages of the Current Data Model

The current flat table model provides several advantages:

- **Simplicity:** The data is easy to understand because all columns are stored in one table.
- **Easy Querying:** Most analysis can be performed without joins.
- **Fast Exploration:** Analysts can quickly group and filter data by category, region, customer segment, or product.
- **Portfolio Friendly:** The structure is suitable for demonstrating SQL querying and business analysis skills.
- **Direct KPI Calculation:** Metrics such as total sales, total profit, quantity sold, and total orders can be calculated directly.

8.10 Limitations of the Current Data Model

Although the flat table model is useful for this project, it also has some limitations:

- **Data Redundancy:** Customer, product, and geography details may repeat across multiple rows.
- **Limited Scalability:** A single-table structure may become inefficient for very large datasets.
- **Maintenance Challenges:** Updating repeated values can be difficult in larger systems.
- **No Separate Dimensions:** Customer, product, geography, and date details are not stored as separate dimension tables.
- **Less Suitable for Enterprise Reporting:** Large business intelligence systems usually require more structured data models.

These limitations do not prevent analysis in this project, but they show how the database could be improved for a production-level analytics system.

8.11 Future Conversion into a Star Schema

In the future, the flat table could be converted into a **Star Schema** to support scalable business intelligence reporting. A Star Schema organizes data into one central fact table and multiple dimension tables.

The central fact table could store measurable transaction values such as sales, quantity, discount, and profit. Dimension tables could store descriptive information related to customers, products, geography, shipping, and dates.

A possible future Star Schema design could include:

- **FactSales:** Stores transaction-level measures such as Sales, Quantity, Discount, Profit, Order ID, Product ID, Customer ID, Date Key, and Geography Key.
- **DimCustomer:** Stores Customer ID, Customer Name, and Segment.
- **DimProduct:** Stores Product ID, Product Name, Category, and Sub-Category.
- **DimGeography:** Stores Country, Region, State, City, and Postal Code.
- **DimDate:** Stores Order Date, Year, Quarter, Month, and Day.
- **DimShipping:** Stores Ship Mode and shipping-related attributes.

Converting the dataset into a Star Schema would reduce redundancy, improve reporting performance, and make the database more suitable for dashboarding tools such as Power BI, Tableau, and Looker Studio.

8.12 Conclusion

The database schema used in this project follows a flat analytical table design. This structure is appropriate for SQL-based portfolio analysis because it is simple, readable, and effective for direct querying. The **superstore_sales** table supports analysis across products, customers, regions, cities, shipping modes, and time periods.

While the current design is suitable for project-level analysis, future improvements could include converting the dataset into a Star Schema. This would make the database more scalable, reduce redundancy, and support enterprise-style business intelligence reporting.

9 Data Validation and Cleaning

Data validation was performed before analysis to ensure the reliability of results. This step is important because inaccurate, missing, or duplicate data can lead to misleading business conclusions.

Table 9.1: Data Validation Results

Validation Check	Result	Status
Total row count	9,694 rows	Passed
Total column count	21 columns	Passed
Missing value check	No missing values	Passed
Duplicate Row ID check	No duplicate Row IDs	Passed
Date range check	2014–2017	Passed
Cleaning status	Cleaned and validated	Passed

9.1 Validation Insight

The dataset was suitable for analysis because it did not contain missing values or duplicate Row IDs. This improves confidence in the accuracy of KPIs, rankings, and business comparisons. Since the dataset was cleaned and validated before analysis, the resulting insights can be considered dependable for portfolio-level business reporting.

10 KPI Analysis

KPI analysis provides a high-level summary of the overall business performance. These metrics help stakeholders quickly understand the scale and profitability of the business.

Table 10.1: Overall Business KPIs

KPI	Value
Total Sales	$2,272,449.86
Total Profit	$286,857.75
Total Quantity Sold	36,749
Total Orders	4,931

10.1 Business Insight

The business generated more than **$2.27 million** in total sales and approximately **$286.86 thousand** in total profit. This indicates that the overall business is profitable. However, the gap between sales and profit shows that revenue alone is not enough to judge performance. A business can generate high sales but still face profitability issues if discounts, costs, or loss-making products are not controlled.

The total order count of **4,931** and quantity sold of **36,749** indicate a strong transactional base. These metrics provide a foundation for deeper analysis across products, regions, customer segments, and shipping methods.

11 Sample SQL Queries Used in the Project

This chapter presents sample SQL queries used in the project **"SQL Sales Performance Analysis Using MySQL"**. These queries were written and executed in MySQL Workbench using the database **sales_analysis** and the table **superstore_sales**. The queries cover database selection, table verification, KPI calculation, product analysis, category analysis, regional analysis, customer analysis, shipping analysis, and year-wise trend analysis.

11.1 Query 1: Database Selection

Query Objective: To select the project database before performing analysis.

```
USE sales_analysis;
```

Business Purpose: Selecting the correct database ensures that all SQL queries are executed on the intended project dataset. This prevents errors and keeps the analysis organized within the dedicated database environment.

11.2 Query 2: Table Verification

Query Objective: To verify that the required table exists and view sample records from the dataset.

```
SELECT *
FROM superstore_sales
LIMIT 10;
```

Business Purpose: This query helps confirm that the dataset was imported correctly into MySQL. Viewing sample records allows the analyst to understand the structure of the table and check whether important columns such as sales, profit, category, region, and order date are available.

11.3 Query 3: Total Sales Calculation

Query Objective: To calculate the total sales generated by the business.

```
SELECT
    ROUND(SUM(Sales), 2) AS total_sales
FROM superstore_sales;
```

Business Purpose: Total sales is one of the most important business KPIs. It shows the overall revenue generated during the analysis period and helps measure the size of business operations.

11.4 Query 4: Total Profit Calculation

Query Objective: To calculate the total profit earned by the business.

```
SELECT
    ROUND(SUM(Profit), 2) AS total_profit
FROM superstore_sales;
```

Business Purpose: Total profit helps evaluate whether the business is financially successful. While sales show revenue, profit shows the actual earning after considering costs and losses.

11.5 Query 5: Total Quantity Calculation

Query Objective: To calculate the total quantity of products sold.

```
SELECT
    SUM(Quantity) AS total_quantity_sold
FROM superstore_sales;
```

Business Purpose: Total quantity sold helps measure product movement and customer demand. It is useful for understanding sales volume, inventory planning, and operational scale.

11.6 Query 6: Total Orders Calculation

Query Objective: To calculate the total number of unique orders.

```
SELECT
    COUNT(DISTINCT ‘Order ID‘) AS total_orders
FROM superstore_sales;
```

Business Purpose: Total orders show how many separate customer purchases were made. This metric helps understand transaction volume and supports further analysis such as average order value.

11.7 Query 7: Product Profitability Analysis

Query Objective: To identify the top profitable products based on total profit.

```
SELECT
    ‘Product Name‘,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit
FROM superstore_sales
GROUP BY ‘Product Name‘
ORDER BY total_profit DESC
LIMIT 10;
```

Business Purpose: This query identifies products that contribute the most to business profit. It helps management focus on high-value products for promotion, inventory planning, and strategic growth.

11.8 Query 8: Loss-Making Products

Query Objective: To identify products that generated negative profit.

```
SELECT
    `Product Name`,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_loss
FROM superstore_sales
GROUP BY `Product Name`
HAVING total_loss < 0
ORDER BY total_loss ASC
LIMIT 10;
```

Business Purpose: This query highlights products that are causing financial losses. These products should be reviewed for pricing, discounts, procurement cost, shipping cost, or possible discontinuation.

11.9 Query 9: Category Analysis

Query Objective: To analyze sales and profit performance by product category.

```
SELECT
    Category,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit,
    SUM(Quantity) AS total_quantity
FROM superstore_sales
GROUP BY Category
ORDER BY total_profit DESC;
```

Business Purpose: Category analysis helps compare the performance of major product groups. It allows the business to identify which categories are strong profit drivers and which categories need improvement.

11.10 Query 10: Sub-Category Analysis

Query Objective: To analyze sales and profit performance by product sub-category.

```
SELECT
    `Sub-Category`,
```

```
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit,
    SUM(Quantity) AS total_quantity
FROM superstore_sales
GROUP BY ‘Sub-Category‘
ORDER BY total_profit DESC;
```

Business Purpose: Sub-category analysis provides deeper insight than category-level analysis. It helps identify specific product groups that are profitable or loss-making within broader categories.

11.11 Query 11: Region Analysis

Query Objective: To analyze sales and profit by region.

```
SELECT
    Region,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit,
    COUNT(DISTINCT ‘Order ID‘) AS total_orders
FROM superstore_sales
GROUP BY Region
ORDER BY total_profit DESC;
```

Business Purpose: Region analysis helps identify geographic areas that perform well or underperform. This supports regional planning, sales strategy, and resource allocation.

11.12 Query 12: City Analysis

Query Objective: To analyze city-wise sales and profit performance.

```
SELECT
    City,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit,
    COUNT(DISTINCT ‘Order ID‘) AS total_orders
FROM superstore_sales
```

```
GROUP BY City
ORDER BY total_profit DESC
LIMIT 10;
```

Business Purpose: City-level analysis helps identify local markets that generate strong profit. It also supports deeper investigation into cities with high sales but low or negative profitability.

11.13 Query 13: Customer Segment Analysis

Query Objective: To analyze sales and profit by customer segment.

```
SELECT
    Segment,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit,
    COUNT(DISTINCT ‘Order ID‘) AS total_orders
FROM superstore_sales
GROUP BY Segment
ORDER BY total_profit DESC;
```

Business Purpose: Customer segment analysis shows which customer groups contribute the most to revenue and profit. This helps design better marketing, retention, and sales strategies for each segment.

11.14 Query 14: Ship Mode Analysis

Query Objective: To analyze business performance by shipping mode.

```
SELECT
    ‘Ship Mode‘,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit,
    COUNT(DISTINCT ‘Order ID‘) AS total_orders
FROM superstore_sales
GROUP BY ‘Ship Mode‘
ORDER BY total_profit DESC;
```

Business Purpose: Ship mode analysis helps evaluate how different delivery methods contribute to business performance. It supports decisions related to shipping strategy, delivery cost control, and customer service planning.

11.15 Query 15: Year-Wise Trend Analysis

Query Objective: To analyze year-wise sales and profit trends.

```
SELECT
    YEAR(‘Order Date‘) AS order_year,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit,
    SUM(Quantity) AS total_quantity,
    COUNT(DISTINCT ‘Order ID‘) AS total_orders
FROM superstore_sales
GROUP BY YEAR(‘Order Date‘)
ORDER BY order_year;
```

Business Purpose: Year-wise trend analysis helps understand business growth over time. It shows whether sales and profit are increasing, decreasing, or fluctuating across years, which supports long-term planning and performance evaluation.

12 Product Analysis

Product analysis identifies which products contribute most to profit and which products generate the highest losses. This is important because product-level decisions directly affect revenue, margin, inventory planning, and marketing strategy.

12.1 Top Profitable Products

Table 12.1: Top 5 Profitable Products

Rank	Product Name	Profit
1	Canon imageCLASS 2200 Advanced Copier	$25,199.93
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine	$7,753.04
3	Hewlett Packard LaserJet 3310 Copier	$6,983.88
4	Canon PC1060 Personal Laser Copier	$4,570.93
5	HP Designjet T520 Inkjet Printer	$4,094.98

12.2 Top Loss-Making Products

Table 12.2: Top 5 Loss-Making Products

Rank	Product Name	Profit / Loss
1	Cubify CubeX 3D Printer Double Head Print	-$8,879.97
2	Lexmark MX611dhe Monochrome Laser Printer	-$4,589.97
3	Cubify CubeX 3D Printer Triple Head Print	-$3,839.99
4	Chromcraft Bull-Nose Wood Oval Conference Table	-$2,876.12
5	Bush Advantage Collection Racetrack Conference Table	-$1,934.40

12.3 Business Insight

The top profitable products are mainly technology-related products, especially copiers and printers. The Canon imageCLASS 2200 Advanced Copier is the strongest product, generating **$25,199.93** in profit. This indicates that selected high-value technology products can strongly improve profitability.

At the same time, the loss-making product list also includes technology products such as 3D printers and laser printers. This means the Technology category has both high opportunity and high risk. High-ticket products require careful pricing, discount control, and demand analysis because a small number of unprofitable transactions can create large losses.

13 Category Analysis

Category analysis compares sales and profit across the three main product categories.

Table 13.1: Category Performance

Category	**Sales**	**Profit**
Technology	\$835,900.07	\$145,387.10
Furniture	\$733,046.86	\$16,980.77
Office Supplies	\$703,502.93	\$120,489.89

13.1 Business Insight

Technology is the strongest category, generating the highest sales and the highest profit. Office Supplies generated slightly lower sales than Furniture but produced much higher profit. Furniture generated high sales but very low profit compared with the other categories.

This shows that sales volume does not always represent business success. Furniture appears to have margin issues, possibly because of discounting, shipping costs, procurement costs, or loss-making sub-categories such as tables and bookcases.

14 Sub-Category Analysis

Sub-category analysis provides deeper understanding of profitability beneath the category level.

Table 14.1: Top Profitable Sub-Categories

Rank	**Sub-Category**
1	Copiers
2	Phones
3	Accessories
4	Paper
5	Binders

Table 14.2: Loss-Making Sub-Categories

Rank	**Sub-Category**
1	Tables
2	Bookcases
3	Supplies

14.1 Business Insight

Copiers, phones, accessories, paper, and binders are the most profitable sub-categories. These sub-categories should be treated as important profit drivers. Tables, bookcases, and supplies are loss-making sub-categories. The presence of tables and bookcases in the loss-making group explains why Furniture has weak profitability despite strong sales.

15 Region Analysis

Regional analysis compares sales and profit across the West, East, Central, and South regions.

Table 15.1: Regional Sales Performance

Region	Sales
West	$713,471.34
East	$672,194.05
Central	$497,800.87
South	$388,983.59

Table 15.2: Regional Profit Performance

Region	Profit
West	$106,021.15
East	$90,672.01
South	$46,035.69
Central	$40,128.90

15.1 Business Insight

The West region is the strongest region in both sales and profit. The East region is also a strong performer and ranks second in both metrics. The Central region has higher sales than the South region but lower profit, which indicates weaker profitability. This may be due to higher discounts, less profitable products, or loss-making cities within the region.

16 State Analysis

State analysis helps identify which states contribute most to profitability.

Table 16.1: Top Profitable States

Rank	State
1	California
2	New York
3	Washington
4	Michigan
5	Virginia

16.1 Business Insight

California, New York, and Washington are the strongest profit-generating states. These states should be considered key markets for customer retention, promotional planning, and revenue expansion. Michigan and Virginia also show strong profitability and may offer opportunities for further growth.

17 City Analysis

City analysis gives a more detailed view of geographic performance and helps identify local markets that are profitable or loss-making.

Table 17.1: Top Sales Cities

Rank	City
1	New York City
2	Los Angeles
3	Seattle
4	San Francisco
5	Philadelphia

Table 17.2: Top Profit Cities

Rank	City
1	New York City
2	Los Angeles
3	Seattle
4	San Francisco
5	Detroit

Table 17.3: Loss-Making Cities

Rank	City
1	Philadelphia
2	Houston
3	San Antonio
4	Lancaster
5	Chicago

17.1 Business Insight

New York City, Los Angeles, Seattle, and San Francisco are strong cities because they appear among the top cities for both sales and profit. Philadelphia is an important concern because it appears in the top sales cities but is also the highest loss-making city. This means that high sales in Philadelphia are not converting into profit.

The loss-making cities should be reviewed carefully. Possible reasons may include excessive discounts, unprofitable product mix, high shipping costs, or low-margin customer behavior.

18 Customer Segment Analysis

Customer segment analysis compares business contribution across Consumer, Corporate, and Home Office customers.

Table 18.1: Customer Segment Performance

Segment	**Sales**	**Profit**
Consumer	$1,150,166.18	$132,669.78
Corporate	$696,604.51	$90,366.30
Home Office	$425,679.16	$59,821.68

18.1 Business Insight

The Consumer segment is the largest contributor to both sales and profit. Corporate customers rank second, while Home Office customers contribute the lowest sales and profit. Since all three segments are profitable, the business has a healthy customer base across different customer types.

The Consumer segment should remain a priority for growth campaigns. Corporate customers can be developed through account-based strategies, while the Home Office segment can be targeted with bundles and remote-work product offers.

19 Ship Mode Analysis

Ship mode analysis evaluates performance across different delivery methods.

Table 19.1: Ship Mode Performance

Ship Mode	Sales	Profit
Standard Class	$1,342,260.19	$161,547.78
Second Class	$453,341.85	$56,505.74
First Class	$349,494.85	$48,778.85
Same Day	$127,352.97	$16,025.38

19.1 Business Insight

Standard Class is the strongest shipping mode by both sales and profit. This suggests that most customers either prefer Standard Class or that it is the most commonly used shipping option. Same Day shipping has the lowest sales and profit, which may be due to limited demand, higher cost, or operational constraints.

Shipping strategy should be evaluated not only by customer convenience but also by profitability. Faster shipping modes should be monitored to ensure that service costs do not reduce margins.

20 Year-wise Trend Analysis

Year-wise trend analysis helps identify how sales and profit changed from 2014 to 2017.

Table 20.1: Year-Wise Sales and Profit Performance

Year	Sales	Profit
2014	$481,763.80	$49,044.43
2015	$464,426.24	$60,907.69
2016	$601,265.26	$80,130.65
2017	$724,994.56	$92,774.99

20.1 Business Insight

Sales decreased slightly from 2014 to 2015 but increased significantly in 2016 and 2017. Profit increased every year from 2014 to 2017. This is a positive sign because it shows that profitability improved consistently over time, even when sales declined in 2015.

The strongest year was 2017, with sales of **$724,994.56** and profit of **$92,774.99**. The company should study the strategies, product mix, and market conditions that contributed to the 2016 and 2017 growth.

21 Advanced SQL Analysis

This chapter demonstrates advanced SQL concepts used to perform deeper business analysis on the Superstore Sales Dataset. The database used for this project is **sales_analysis**, and the main table used for analysis is **superstore_sales**. Advanced SQL techniques such as `CASE WHEN`, ranking functions, Common Table Expressions, window functions, and subqueries help convert basic summaries into more powerful analytical outputs.

21.1 CASE WHEN: Profit Categorization

21.1.1 Concept Explanation

The `CASE WHEN` statement is used to create conditional logic in SQL. It allows the analyst to classify records into different groups based on defined business rules.

21.1.2 Business Use Case

In this project, `CASE WHEN` can be used to categorize products into profit groups such as high profit, moderate profit, low profit, or loss-making products.

21.1.3 SQL Query

```
USE sales_analysis;

SELECT
    ‘Product Name‘,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit,
    CASE
        WHEN SUM(Profit) > 5000 THEN ’High Profit’
```

```
        WHEN SUM(Profit) BETWEEN 1000 AND 5000 THEN 'Moderate Profit'
        WHEN SUM(Profit) BETWEEN 0 AND 999.99 THEN 'Low Profit'
        ELSE 'Loss-Making'
    END AS profit_category
FROM superstore_sales
GROUP BY `Product Name`
ORDER BY total_profit DESC;
```

21.1.4 Output Interpretation

The output classifies each product according to its total profit contribution. Products with very high profit are marked as **High Profit**, while products with negative profit are marked as **Loss-Making**.

21.1.5 Business Insight

This classification helps the business quickly identify which products should be promoted, monitored, improved, or discontinued. High-profit products can be prioritized in marketing and inventory planning, while loss-making products require pricing and discount review.

21.2 RANK(): Product Profit Ranking

21.2.1 Concept Explanation

The `RANK()` function assigns a ranking position to rows based on a specified order. If two records have the same value, they receive the same rank, and the next rank is skipped.

21.2.2 Business Use Case

`RANK()` can be used to rank products by total profit and identify the top-performing products.

21.2.3 SQL Query

```
SELECT
```

```
    `Product Name`,
    ROUND(SUM(Profit), 2) AS total_profit,
    RANK() OVER (ORDER BY SUM(Profit) DESC) AS profit_rank
FROM superstore_sales
GROUP BY `Product Name`
ORDER BY profit_rank
LIMIT 10;
```

21.2.4 Output Interpretation

The output displays the top 10 products based on total profit. The product with the highest total profit receives rank 1.

21.2.5 Business Insight

Product ranking helps identify the strongest profit contributors. These products should be protected through proper stock availability, targeted promotions, and customer-focused sales strategies.

21.3 DENSE_RANK(): Regional Sales Ranking

21.3.1 Concept Explanation

The `DENSE_RANK()` function is similar to `RANK()`, but it does not skip rank numbers when there are ties. This makes it useful when a continuous ranking sequence is required.

21.3.2 Business Use Case

`DENSE_RANK()` can be used to rank regions by total sales without leaving gaps in the ranking sequence.

21.3.3 SQL Query

```
SELECT
    Region,
    ROUND(SUM(Sales), 2) AS total_sales,
```

```
    DENSE_RANK() OVER (ORDER BY SUM(Sales) DESC) AS sales_rank
FROM superstore_sales
GROUP BY Region
ORDER BY sales_rank;
```

21.3.4 Output Interpretation

The output ranks each region according to total sales. The region with the highest sales receives rank 1, followed by the remaining regions in descending order.

21.3.5 Business Insight

Regional ranking helps management compare market performance across geographic areas. Strong regions can be used as benchmarks, while weaker regions may require marketing, pricing, or distribution improvements.

21.4 ROW_NUMBER(): Customer Ranking

21.4.1 Concept Explanation

The `ROW_NUMBER()` function assigns a unique sequential number to each row based on the specified ordering. Unlike `RANK()` and `DENSE_RANK()`, it does not assign the same number to tied records.

21.4.2 Business Use Case

`ROW_NUMBER()` can be used to create a unique customer ranking based on total sales contribution.

21.4.3 SQL Query

```
SELECT
    `Customer Name`,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROW_NUMBER() OVER (ORDER BY SUM(Sales) DESC) AS customer_position
FROM superstore_sales
```

```
GROUP BY ‘Customer Name‘
ORDER BY customer_position
LIMIT 10;
```

21.4.4 Output Interpretation

The output lists the top 10 customers by total sales and assigns each customer a unique position.

21.4.5 Business Insight

Customer ranking helps identify high-value customers. These customers can be targeted for loyalty programs, personalized offers, and relationship management strategies.

21.5 Common Table Expression: Top-N Product Analysis

21.5.1 Concept Explanation

A Common Table Expression, or CTE, is a temporary named result set created using the `WITH` clause. It improves query readability and is useful when a query requires multiple logical steps.

21.5.2 Business Use Case

A CTE can be used to calculate product-level profit first and then filter only the top profitable products.

21.5.3 SQL Query

```
WITH product_profit AS (
    SELECT
        ‘Product Name‘,
        ROUND(SUM(Sales), 2) AS total_sales,
        ROUND(SUM(Profit), 2) AS total_profit
```

```
    FROM superstore_sales
    GROUP BY `Product Name`
)
SELECT
    `Product Name`,
    total_sales,
    total_profit
FROM product_profit
ORDER BY total_profit DESC
LIMIT 5;
```

21.5.4 Output Interpretation

The CTE first calculates total sales and profit for each product. The final query then selects the top 5 products based on total profit.

21.5.5 Business Insight

CTEs make complex analysis easier to understand and maintain. This approach is useful when building reusable analytical logic for product performance reports.

21.6 Window Functions: Running Sales Total by Year

21.6.1 Concept Explanation

Window functions perform calculations across a set of rows related to the current row without collapsing the result into a single grouped output. They are useful for rankings, running totals, moving averages, and cumulative performance analysis.

21.6.2 Business Use Case

A window function can be used to calculate cumulative sales over years and understand business growth over time.

21.6.3 SQL Query

```
WITH yearly_sales AS (
    SELECT
        YEAR(`Order Date`) AS order_year,
        ROUND(SUM(Sales), 2) AS total_sales
    FROM superstore_sales
    GROUP BY YEAR(`Order Date`)
)
SELECT
    order_year,
    total_sales,
    ROUND(
        SUM(total_sales) OVER (
            ORDER BY order_year
            ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW
        ), 2
    ) AS running_total_sales
FROM yearly_sales
ORDER BY order_year;
```

21.6.4 Output Interpretation

The output shows total sales for each year along with cumulative sales up to that year. For example, the running total for 2016 includes sales from 2014, 2015, and 2016.

21.6.5 Business Insight

Running totals help track cumulative business growth. This is useful for long-term performance monitoring and understanding how revenue builds over time.

21.7 Subqueries: Products Above Average Sales

21.7.1 Concept Explanation

A subquery is a query written inside another SQL query. Subqueries are useful when one calculation depends on the result of another calculation.

21.7.2 Business Use Case

A subquery can be used to identify products whose total sales are higher than the average product-level sales.

21.7.3 SQL Query

```
SELECT
    product_summary.‘Product Name‘,
    product_summary.total_sales,
    product_summary.total_profit
FROM (
    SELECT
        ‘Product Name‘,
        ROUND(SUM(Sales), 2) AS total_sales,
        ROUND(SUM(Profit), 2) AS total_profit
    FROM superstore_sales
    GROUP BY ‘Product Name‘
) AS product_summary
WHERE product_summary.total_sales > (
    SELECT AVG(product_sales.total_sales)
    FROM (
        SELECT
            ‘Product Name‘,
            SUM(Sales) AS total_sales
        FROM superstore_sales
        GROUP BY ‘Product Name‘
    ) AS product_sales
)
ORDER BY product_summary.total_sales DESC;
```

21.7.4 Output Interpretation

The output returns products whose total sales are greater than the average sales value across all products.

21.7.5 Business Insight

This analysis helps identify products with above-average sales performance. These products may represent strong demand and can be prioritized for inventory planning, promotional campaigns, and sales forecasting.

21.8 Additional Advanced Query: Sales Segmentation Using CASE WHEN

21.8.1 Concept Explanation

Sales segmentation divides records into groups based on sales value. This can be done using `CASE WHEN` to classify orders as high-value, medium-value, or low-value.

21.8.2 Business Use Case

This analysis helps understand how many transactions belong to each sales value group.

21.8.3 SQL Query

```
SELECT
    CASE
        WHEN Sales >= 1000 THEN 'High-Value Sale'
        WHEN Sales BETWEEN 250 AND 999.99 THEN 'Medium-Value Sale'
        ELSE 'Low-Value Sale'
    END AS sales_segment,
    COUNT(*) AS number_of_transactions,
    ROUND(SUM(Sales), 2) AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit
FROM superstore_sales
GROUP BY sales_segment
ORDER BY total_sales DESC;
```

21.8.4 Output Interpretation

The output groups transactions into high-value, medium-value, and low-value sales segments. It also shows the number of transactions, total sales, and total profit for each segment.

21.8.5 Business Insight

Sales segmentation helps the business understand whether revenue is mainly driven by a small number of high-value transactions or a large number of smaller transactions. This can support pricing, marketing, and customer targeting strategies.

21.9 Additional Advanced Query: Top Product in Each Region

21.9.1 Concept Explanation

Window functions can be combined with CTEs to perform group-wise ranking. This is useful when the analyst needs to identify the top record within each group.

21.9.2 Business Use Case

This query identifies the most profitable product in each region.

21.9.3 SQL Query

```
WITH regional_product_profit AS (
    SELECT
        Region,
        ‘Product Name‘,
        ROUND(SUM(Profit), 2) AS total_profit
    FROM superstore_sales
    GROUP BY Region, ‘Product Name‘
),
ranked_products AS (
    SELECT
```

```
        Region,
        `Product Name`,
        total_profit,
        RANK() OVER (
            PARTITION BY Region
            ORDER BY total_profit DESC
        ) AS regional_profit_rank
    FROM regional_product_profit
)
SELECT
    Region,
    `Product Name`,
    total_profit
FROM ranked_products
WHERE regional_profit_rank = 1
ORDER BY Region;
```

21.9.4 Output Interpretation

The output shows the most profitable product for each region. The ranking is calculated separately within each region using `PARTITION BY Region`.

21.9.5 Business Insight

This analysis helps identify region-specific profit drivers. Regional managers can use this information to promote products that perform best in their local markets.

22 Business Insights

The complete SQL analysis produced the following key business insights:

1. The overall business is profitable, with total profit of **$286,857.75**.
2. Total sales of **$2,272,449.86** show strong revenue generation across the dataset.
3. Technology is the strongest category in both sales and profit.
4. Office Supplies is more profitable than Furniture despite generating lower sales.
5. Furniture has weak profitability and requires deeper review.
6. Copiers are a major profit driver within the business.
7. Tables, bookcases, and supplies are loss-making sub-categories.
8. The West region is the top-performing region in both sales and profit.
9. The Central region has weaker profitability compared with its sales volume.
10. New York City, Los Angeles, Seattle, and San Francisco are strong city markets.
11. Philadelphia is a major concern because it has high sales but also high losses.
12. Consumer customers generate the highest sales and profit.
13. Standard Class shipping contributes the highest sales and profit.
14. Profit improved every year from 2014 to 2017, showing positive business progress.

23 Business Recommendations

Based on the analysis, the following recommendations are proposed:

23.1 Product Recommendations

- Increase focus on high-profit products such as Canon copiers and other profitable Technology products.
- Review pricing and discount policies for loss-making products, especially 3D printers and conference tables.
- Identify whether loss-making products should be repriced, bundled, restricted, or discontinued.
- Track product-level profit regularly to prevent repeated losses.

23.2 Category and Sub-Category Recommendations

- Treat Technology and Office Supplies as key profit-generating categories.
- Investigate Furniture profitability, especially tables and bookcases.
- Reduce unnecessary discounts in weak-margin categories.
- Improve procurement and shipping cost control for Furniture products.

23.3 Regional and Geographic Recommendations

- Use the West region as a benchmark for strong sales and profit performance.
- Study the East region for additional expansion opportunities.

- Review the Central region to identify why profit is lower despite moderate sales.
- Investigate loss-making cities such as Philadelphia, Houston, San Antonio, Lancaster, and Chicago.

23.4 Customer and Shipping Recommendations

- Continue targeting the Consumer segment because it generates the highest sales and profit.
- Build account-based campaigns for Corporate customers.
- Create special offers for Home Office customers using bundled product strategies.
- Maintain Standard Class as the primary shipping mode while monitoring faster shipping options for profitability.

24 Conclusion

This project successfully demonstrates how SQL can be used to perform end-to-end business analysis on retail sales data. Using MySQL Workbench and the Superstore Sales Dataset, the project analyzed overall KPIs, product performance, category performance, sub-category profitability, regional results, state and city-level performance, customer segments, shipping modes, and year-wise trends.

The analysis showed that the business generated total sales of **$2,272,449.86** and total profit of **$286,857.75**. Technology was the most profitable category, while Furniture showed weak profitability. The West region was the strongest region, and the Consumer segment was the largest customer segment. The analysis also identified important loss areas such as specific products, sub-categories, and cities.

Overall, this project highlights the practical value of SQL in data analytics. It shows how raw transactional data can be transformed into meaningful insights and recommendations that support business decisions. For a data analyst portfolio, this project demonstrates technical SQL skills, analytical thinking, business understanding, and professional reporting ability.

25 Future Scope

The project can be extended in the following ways:

- Create an interactive dashboard using Power BI, Tableau, or Looker Studio.
- Perform monthly and quarterly sales trend analysis.
- Analyze discount impact on profit and margin.
- Build customer retention and repeat purchase analysis.
- Create product profitability scorecards.
- Develop sales forecasting models using machine learning.
- Segment customers based on purchase behavior.
- Convert the flat table into a star schema with fact and dimension tables.
- Automate SQL reports using stored procedures or scheduled scripts.

26 References

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5. Business intelligence concepts related to KPI reporting, profitability analysis, customer segmentation, and sales performance evaluation.